Muhammad Nadeem

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Professional Summary

AI Research Engineer working on **LLM/VLM and vision models** end to end — agentic RAG, lightweight fine-tuning with Hugging Face PEFT (LoRA/QLoRA), and inference optimization via quantization, ONNX, TensorRT, OpenVINO, and vLLM/Ollama for constrained GPUs and edge devices. M.S. in Computer Science (Sejong University, CVPR Lab); **13 publications** (5 first-author), including WACV 2026 (LLVM-AD workshop) and ACPR 2025; Best Paper Award, ICNGC 2023.

Education

2nd Mar-22 – 18th Feb-24 — **Sejong University**, Seoul, South Korea
- M.S. in Computer Science and Engineering
- Thesis: FlameNet: Efficient Deep Learning for Fire Detection in Smart Cities
- GPA: 4.25/4.5, Presentation Link

1st Feb-14 – 23rd Apr-18 — **The Islamia University of Bahawalpur**, Bahawalpur, Pakistan
- B.S. in Computer Science
- GPA: 3.28/4.0

Technical Skills

LLM / VLM Qwen2.5-VL, HyperCLOVA X SEED, CLIP dual encoders, VQA, zero-shot cross-modal recognition

Fine-tuning Hugging Face Transformers & PEFT, LoRA/QLoRA, instruction tuning, chain-of-thought prompting

RAG LangChain, hybrid retrieval (dense + BM25 + knowledge graph), BGE-M3, RRF, cross-encoder reranking

Inference Opt. Quantization (INT8/4-bit, PTQ & QAT), ONNX Runtime, TensorRT, OpenVINO, TFLite, distillation

Serving vLLM, Ollama, GGUF (llama.cpp), FastAPI, Flask, REST APIs, RTSP ingestion, React dashboards

Detection YOLOv8/v12/v26, Faster R-CNN, DETR, SSD, YOLOX, YOLO-NAS, GroundingDINO, MobileNet

Segmentation Mask R-CNN instance segmentation, semantic segmentation, OCR pipelines

Tracking / Pose Deep SORT, YOLO-pose 17-keypoint skeletons, fall detection, face de-identification

Generative FLUX.2 [dev], Stable Diffusion, GANs, `diffusers`, synthetic data generation

Training PyTorch, TensorFlow, OpenCV, multi-GPU (DP, DDP, multi-node), transfer learning

GPU / Edge HW NVIDIA GPU servers, Jetson Orin Nano, Raspberry Pi 4/5, Intel N100 edge PC

Time-Series N-BEATS, N-HiTS, forecasting and prediction pipelines

Lang & Tools Python, C++, Java, C#, Git, JIRA, Agile/Scrum; Ubuntu 24.04, Windows 11, macOS

Selected Projects (Portfolio)

Full portfolio with demos and code: mnadeemdev.github.io | github.com/MNadeemDev

2026 — **GaN Fab Process-Traceability RAG System** (on-premise LLM + VLM)
Stack: HyperCLOVA X SEED, Qwen2.5-VL 7B, BGE-M3, HF PEFT, Ollama, llama.cpp/GGUF

- On-premise Korean RAG over 14,024 chunks: 12,890 equipment records + a 314-page MES manual.
- Hybrid retrieval: BGE-M3 dense + Korean-bigram BM25 + PageRank knowledge graph, fused by RRF.
- Cross-encoder reranks top-30 to top-5; every answer returned with citations.
- Qwen2.5-VL 7B captions manual screenshots, making in-image content retrievable as text.
- Domain-adapted with LoRA and 4-bit QLoRA; adapters merged to GGUF for Ollama serving.
- 5-step chain-of-thought RCA prompt; auto-generates cited traceability and RCA reports.
- Benchmarked candidate LLMs on an in-house RCA benchmark to select the deployed model.

2025 **SpectroFire — Zero-Shot EO/IR Vision-Language Recognition** (ACPR 2025, ICOIN 2026)
2026 **Stack:** CLIP-style dual encoder, contrastive learning, EO/IR fusion, PyTorch, Jetson

- CLIP-style dual encoder over EO + thermal-IR imagery for zero-shot recognition.
- Classifies by cosine similarity to prompt embeddings; new classes need no retraining.
- Added cross-spectral registration, temporal sync, and density-based geolocation.
- Auto CUDA/CPU device selection for Jetson; edge inference with IoT data networking.

2026 **PA-Drive — End-to-End Autonomous Driving Planner** (2026 AD AI Challenge)
Stack: Transformer decoder, VAD/VADv2 scene encoder, nuScenes, FP16 autocast, Docker

- 6-camera planner: scene encoder to K trajectory candidates via a Transformer decoder.
- Differentiable physical objective (collision, lane, dynamics) then hard-constraint filtering.
- Cut nuScenes planning L2 from **1.902 to 0.246** across four iterations.
- Removed temporal data leakage by switching to official nuScenes splits.
- FP16 autocast with an explicit FLOPs budget check for on-vehicle latency.
- Related first-author paper accepted at **WACV 2026** (LLVM-AD workshop).

2026 **Manhole Worker Safety — Pose-Based Activity Tracking** (ICRA 2027 submission)
Stack: YOLOv8/26-pose, OpenCV, RTSP, ONNX Runtime, OpenVINO, FastAPI, Intel N100

- Worker activity tracking: 17-keypoint pose skeletons tracked from live RTSP camera feeds.
- Fall detection from pose geometry: torso tilt $>55°$ or box aspect >1.10, held 5 frames.
- Dual-track de-identification: blurred face with visible skeleton live, solid mask for storage.
- Face region from keypoints with a shoulder fallback, so no face is left unmasked.
- Per-frame audit log of masking and fall alarms as de-identification compliance evidence.
- Fused with a five-gas sensor array into a four-level risk lattice, decided at the edge.
- Benchmarked ONNX Runtime and OpenVINO latency (p50/p90/p99) on an Intel N100 board.

2026 **Drone Safety-Helmet Compliance Detection with Generative Data Synthesis**
Stack: FLUX.2 [dev] via Hugging Face `diffusers`, YOLO, PyTorch, bfloat16, CPU offload

- Aerial safety-helmet compliance detection for construction sites and ski resorts.
- Synthesized training data with FLUX.2 [dev] image-conditioned generation from a reference.
- Randomized prompts over backgrounds, weather, and poses; 1024×1024 at $\sim$10 m altitude.
- Ran diffusion in bfloat16 with model CPU offload to fit consumer-grade GPUs.
- Augmented a real YOLO-format helmet dataset with the synthetic imagery.

2024 – Present **Real-Time Detection and Tracking on CCTV and Drone Streams**
Stack: YOLOv8/v12/v26, Deep SORT, RTSP/OpenCV, TensorRT, FastAPI, Jetson Orin Nano

- Inference pipelines on live RTSP streams from IP/CCTV cameras and drone feeds.
- Handled decoding, frame sampling, and reconnection for continuous operation.
- Detection and multi-object tracking: debris, sea-water waste, fire and smoke, persons.
- Tuned tiny-object detection and optimized with TensorRT for real-time GPU/edge throughput.
- Served as REST APIs with operator visualization dashboards.

2022 **FlameNet — Lightweight Fire Detection for Smart Cities** (M.S. Thesis)
2024 **Stack:** PyTorch, efficient CNN design, model compression, IoT/edge deployment

- Efficient CNN for fire detection and classification in IoT environments.
- Optimized for low parameter count and inference cost rather than raw accuracy.
- Published in *Smart Cities* (IF 7.0, Q1); related work won Best Paper at ICNGC 2023.

2022 **Fake QR Code Recognition with a Lightweight MobileNet Classifier**
Stack: MobileNet, PyTorch, transfer learning, lightweight CNN design

- Classifier separating authentic from forged QR codes for counterfeit detection.
- Chose MobileNet for low parameter count so verification runs on phones, not servers.
- Trained by transfer learning; tuned the accuracy versus model-size trade-off.

2022 **VPBR — Pumpkin Biophysical Property Measurement with Mask R-CNN**
2023 **Stack:** Mask R-CNN instance segmentation, PyTorch, OpenCV, geometric measurement

- Mask R-CNN segments individual pumpkins; masks converted to length measurements.
- Automatic, low-cost pipeline replacing manual measurement in the field.
- Handled instance separation under occlusion and varied viewpoints.
- Published in *Plants* (IF 4.5, Q1) as VPBR.

Selected Publications

13 publications; 5 as first author. Full list: [Google Scholar]

Vision-Language Models, Lightweight Models and Edge Inference (2025–2026)

WACV 2026 **Workshop** — **Lightweight Multi-Scale Fusion for Real-Time Autonomous Driving Segmentation**
Muhammad Nadeem, Hyeyoung Lee
[IEEE/CVF WACV 2026 Workshops — 5th Workshop on Large Language and Vision Models for Autonomous Driving (LLVM-AD), pp. 1721–1728]

ACPR 2025 — **Bringing CLIP to the Edge: A Lightweight Fire Detection System for Real-Time Monitoring with UAV**
Hyeyoung Lee, Muhammad Nadeem
[Asian Conference on Pattern Recognition (ACPR), 2025]

ICRCV 2025 — **A Label-Free Lightweight Prompt-Driven Cross-Modal Fire Detection on Robotic Edge Platforms**
Hyeyoung Lee, Muhammad Nadeem
[The 7th International Conference on Robotics and Computer Vision (ICRCV), 2025]

ICOIN 2026 — **SpectroFire: Zero-Shot Fire and Smoke Detection via Real-Time Edge Inference and IoT Data Networking**
Hyeyoung Lee, Muhammad Nadeem
[The 40th International Conference on Information Networking (ICOIN), 2026]

ICARCA 2025 — **EdgeFlame: A Physics-Inspired Framework for Zero-Shot Fire and Smoke Recognition**
Hyeyoung Lee, Muhammad Nadeem
[The 4th International Conference on Automation, Robotics and Computer Applications (ICARCA), 2025]

Applied AI, IoT and Signal Processing

BigComp 2025 — **IoT Anomaly Detection with Sound-Based Spectral Analysis and Lightweight Model**
Hyeyoung Lee, Muhammad Nadeem
[IEEE International Conference on Big Data and Smart Computing (BigComp), 2025]

BigComp 2026 — **Physics-Consistent Hybrid AI for Quantum Device Manufacturing**
Hyeyoung Lee, Muhammad Nadeem
[IEEE International Conference on Big Data and Smart Computing (BigComp), 2026]

arXiv 2025 — **Spectral Feature Extraction for Robust Network Intrusion Detection Using MFCCs**
Hyeyoung Lee, Muhammad Nadeem, Piotr Tsoi
[arXiv preprint, 2025]

Vision and Deep Learning (M.S. Research, 2023)

Smart Cities **IF 7.0, Q1** — **Visual Intelligence in Smart Cities: A Lightweight Deep Learning Model for Fire Detection in an IoT Environment**
Muhammad Nadeem,Naqqash Dilshad, Norah Saleh Alghamdi, L Minh Dang, Hyoung-Kyu Song, Junyoung Nam, Hyeonjoon Moon
[Journal, PDF]

Plants **IF 4.5, Q1** — **VPBR: An Automatic and Low-Cost Vision-Based Biophysical Properties Recognition Pipeline for Pumpkin**
L. Minh Dang, Muhammad Nadeem, Tan N. Nguyen, Han Yong Park, O New Lee, Hyoung-Kyu Song, Hyeonjoon Moon
[Journal, PDF]

ITCCSCC-2023 — **Machine Learning for Mental Health: A Systematic Study of Seven Approaches for Detecting Mental Disorders**
Muhammad Nadeem, Junaid Rashid, Hyeonjoon Moon, Arailym Dosset
[The 38th International Technical Conference on Circuits/Systems, Computers and Communications, PDF]

ICTC-2023 — **Wheat Diseases Recognition Using Optimal Features Assisted Modified Soft Attention Network**
Muhammad Nadeem, Aqib Khan, Ji-Won Kim, L. Minh Dang, Hyeonjoon Moon
[The 15th International Conference on ICT Convergence]

ICNGC-2023 **Best Paper Award** — **Brest Cancer Recognition Through Visual Intelligence Assisted Lightweight Convolution Neural Network**
Muhammad Nadeem, Haseeb Khan, Wisal Khan, L. Minh Dang, Nguyen Le Quan, Hyeonjoon Moon
[The 9th International Conference on Next Generation Computing (ICNGC 2023)]

Research and Work Experience

15th April-2024 – Present **AI Research Engineer** (LLM/VLM & Computer Vision), Spilab, Seoul, South Korea
Tools: Hugging Face, PEFT/LoRA, LangChain, vLLM, ONNX, TensorRT, OpenVINO, YOLO, FastAPI

- R&D of vision models for detection, segmentation, and tracking on CCTV and drone streams.
- Built agentic RAG over a 14,000-chunk industrial knowledge base returning cited answers.
- Hybrid dense + BM25 + knowledge-graph retrieval with cross-encoder reranking.
- Fine-tuned lightweight LLM/VLMs with Hugging Face PEFT (LoRA/QLoRA) on limited GPU memory.
- Merged adapters and converted to GGUF for local Ollama serving.
- Optimized inference via quantization, ONNX, TensorRT, OpenVINO, and vLLM serving.
- Applied Qwen2.5-VL to document understanding, alongside OCR and VQA pipelines.
- Delivered debris, sea-waste, fire/smoke, and safety-helmet detection projects.
- Pose-based worker activity tracking with fall detection and face de-identification.
- Generated synthetic training data with FLUX.2 [dev] for data-scarce detection tasks.
- Deployed to Jetson Orin Nano and Raspberry Pi via OpenVINO/TFLite conversion.
- Built a production multilingual AI chatbot served through FastAPI.
- Time-series forecasting with N-BEATS and N-HiTS; speech emotion recognition.
- Published 8 papers (2025–2026), including WACV 2026 workshop and ACPR 2025.

15th Mar-22 – 28th Feb-24 **Graduate Researcher** (Computer Vision), CVPR Lab, Sejong University, Seoul, South Korea
Tools: PyTorch, Mask R-CNN, YOLO/Faster R-CNN, Stable Diffusion, multi-GPU training, GANs
- University AI research lab; produced 5 peer-reviewed publications.
- Designed lightweight CNNs for fire detection, medical imaging, and QR forgery detection.
- Mask R-CNN pipelines converting masks to physical measurements for plant phenotyping.
- Scaled training with Data Parallel, DDP, and multi-node GPU clusters.
- Training, fine-tuning, transfer learning, and evaluation of computer vision models.
- Built datasets and applied Stable Diffusion and GANs for augmentation.
- Wrote research papers and technical reports; presented at conferences.

1st Jul-2018 – 30th June-2021 **Software Engineer**, Systems Junction, Lahore, Pakistan
Tools: Java, Android Studio, Retrofit, Room, Glide/Picasso, Firebase, Realm
- Built and maintained production Android applications in Java.
- Optimized computer vision algorithms for performance and memory on mobile devices.
- Developed image processing pipelines and data processing workflows.
- Worked with MVC and singleton patterns, Firebase, Realm, and push notifications.
- Mentored junior engineers and maintained technical documentation.

Honors and Achievements

2023	**Best Paper Award**, ICNGC 2023
2022	**Full Tuition Fee Waiver**, School of Computer Engineering, Sejong University
2022	**Prof. Stipend**, CVPR Lab, Sejong University
2023	**Conference Travel Grant**, Ministry of Education, South Korea
2014	**Merit Scholarship for Bachelor Degree**, The Islamia University of Bahawalpur, Pakistan

Academic Service

Journal Reviewer: IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI), IEEE Trans. on Image Processing (TIP), Fire

Conference Reviewer: 10th Int. Conf. on Next Generation Computing; 40th ITC-CSCC

English Proficiency and Relevant Courseworks

IELTS-AC 6.0	Listening 6.0, Speaking 6.0, Writing 6.0, Reading 5.5
Languages	English (4/5), Korean (1/5)
A+	Introduction to Deep Learning, Advanced Deep Learning, Advanced AI
A+	Computer Vision Topics
A0	Advanced HCI

References

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